

The seven high-level requirements for an open source based IoT ecosystem are:

- The use of open standards
- Taking advantage of community innovation and advances
- Wider integration with platforms and devices
- Accelerated time to market to deliver solutions
- Maintaining control of IoT technology selection and licensing
- Iterating more quickly with less risk
- Laying the foundation for future systems and for scaling

Organisations adopting proprietary IoT platforms are finding themselves tied to limited functionality, locked into working with a particular vendor, and are now thinking of other options. Many of them are looking to open source software as an alternative, recognising the value of the community as a hub for innovation and continuous development for IoT.

In the coming years, the IoT market will move towards adopting an open source and open standards ecosystem, which enables collaborative innovation, and promotes interoperability, flexibility and modularity.

Why should open source software be used for IoT?

IoT promotes the adoption of different open source technologies, standards and protocols that help devices to communicate with one another. The following drivers prompt organisations to adopt open source technologies for IoT.

1. **Cost:** Adopting an open source technology IoT framework leads to minimal investments as it is free for use.
2. **Innovation:** Software developed by the open source community builds newer applications, which leads to better innovation and agility. Developers will be able to build different products, which will be interoperable across different OSs such as Android, Windows, iOS and Linux.
3. **Open API:** Usage of open source APIs for the IoT framework offers a common interface for different software, hardware and systems to communicate with one another.
4. **Libraries:** An open source IoT framework offers a wide range of libraries, SDKs and open source hardware like Raspberry Pi and Arduino, ensuring that companies remain on the cutting-edge of technology by using different

open source tools to customise IoT platforms.

5. **Security:** Open source software can protect data by implementing really strong encryption for the use of the general public (SSH, SSL, PGP, etc), and hence supply the building blocks for mobile security and the protection of data.
6. **Interoperability:** The adoption of open source solves the problem of interoperability.

The open source IoT ecosystem

There is a need for organisations to provide a validated, modular, flexible IoT solution built to be open, interoperable and cost-effective. The solution should deliver end-to-end open source IoT that addresses enterprise level needs.

The characteristics of open source IoT architecture are:

- Loosely coupled, modular and secure
- Platform independent
- Scalable, flexible and can be deployed anywhere
- Based on open standards
- Streaming analytics and machine learning
- Open and interoperable on the hybrid cloud

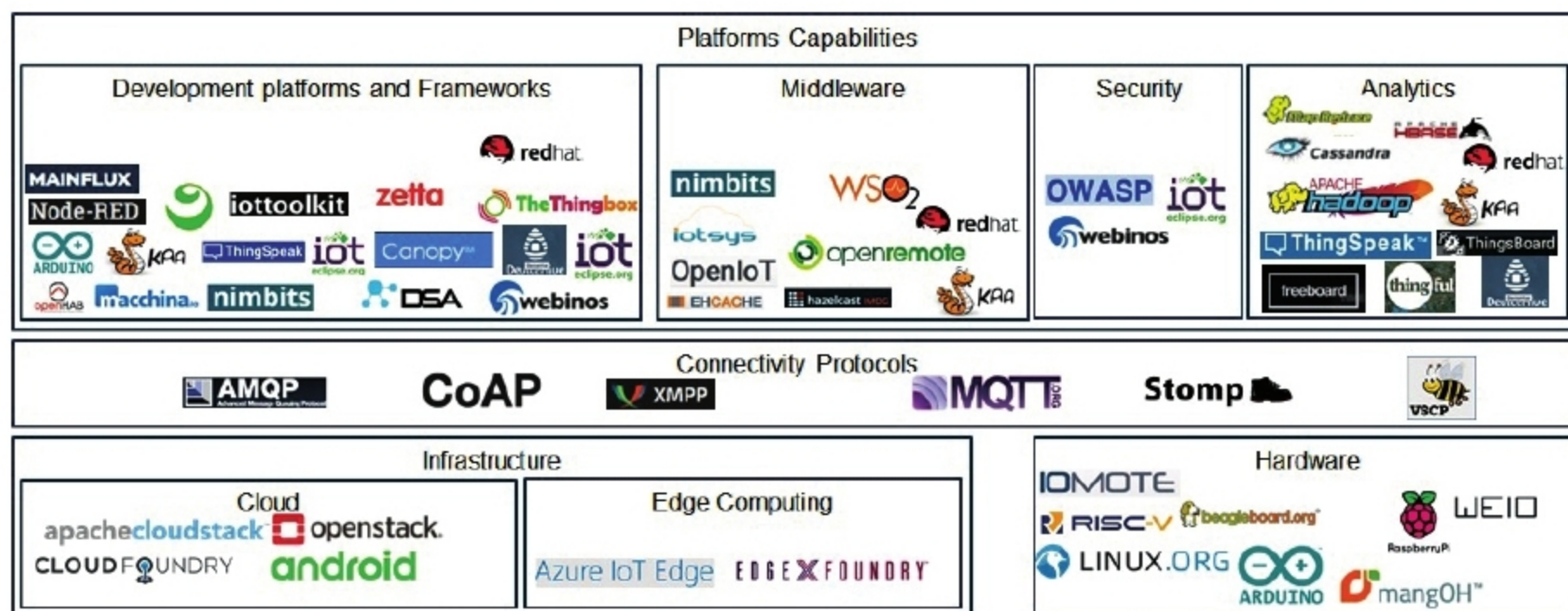


Figure 1: Open source IoT ecosystem